MARVA Report
Job: 0941eb19-0dd0-457e-8d43-fd671cf254a2
Author confirmed: yes
The author has confirmed the references in this report.

Clarity (0)
No finding in this job mapped to this dimension. This dimension is not scored and no finding wa
s invented for it.

Rigor (8)
- Missing selection pressure and co-selection theory for WWTP amplification claim
- Missing environmental fate-transport and QMRA framework to move from gene frequency to health
risk
- Selection-pressure strengthening suggestions include verified but off-topic items
- English-only 2016-2026 review with heterogeneity preventing quantitative synthesis, publicati
on bias risk, and unverified Scopus scope
- Temper WWTP reservoir/amplification and irrigation/drinking-water transmission claims to DNA-
detection limits or add HGT, selection/co-selection, and persistence/expression/HGT framing; ex
plicitly address viability vs extracellular DNA
- Outdated reference: 5. O’neill, J. Tackling Drug-Resistant Infections Globally: Final Report
and Recommendations. News & Publications , 2016.
- Outdated reference: 60. Bradford, P.A. Extended-Spectrum -Lactamases in the 21st Century: Cha
racterization, Epidemiology, and Detection of This Important Resistance Threat. Clin. Microbiol
. Rev. 2001 , 14 , 933 951. [ CrossRef ] [ PubMed ]
- Outdated reference: 148. Rizzo, L.; Manaia, C.; Merlin, C.; Schwartz, T.; Dagot, C.; Ploy, M.
C.; Michael, I.; Fatta-Kassinos, D. Urban wastewater treatment plants as hotspots for antibioti
c resistant bacteria and genes spread into the environment: A review. Sci. Total Environ. 2013
, 447 , 345 360. [ CrossRef ] [ PubMed ]

Novelty (8)
- Missing horizontal gene transfer / mobilome ecology theory to link DNA frequency to dissemina
tion risk
- Three rivals unaddressed: natural background, sink-vs-source/clinical expansion, DNA-persiste
nce/fitness-cost
- Missing theory: Horizontal gene transfer / mobilome ecology theory
- Missing theory: Selection pressure and co-selection theory
- Missing theory: Environmental fate-transport and quantitative microbial risk assessment frame
work
- Rival theory not discussed: Natural / ancient background resistome hypothesis
- Rival theory not discussed: Sink-versus-source / clinical clonal expansion hypothesis
- Rival theory not discussed: Fitness-cost and DNA-persistence / viability-decay hypothesis

References (136)
- 124 ’missing’ numbers are unconfirmed artifact, not proven absent
- Reference strings merged into single blocks preventing reliable verification

- Reference 13 falsely listed as uncited; it is present
- References 3 and 7 flagged as No match
- 10 early references ambiguous and need manual check
- Outdated refs used for fast-moving claims: 5 (2016), 60 (2001), 148 (2013)
- Re-parse full 1-471 reference list and re-run numeric-bracket check; do not report 16-152 as missing until tail is parsed
- Remove false uncited 13; correct uncited list to empty pending re-parse
- Update or justify outdated citations 60 (2001) and 148 (2013) and 5 (2016) where used for current epidemiology/treatment efficacy; cite current systematic reviews for CTX-M/SHV/NDM/OXA
- Missing citation: 16
- Missing citation: 17
- Missing citation: 21
- Missing citation: 22
- Missing citation: 23
- Missing citation: 24
- Missing citation: 25
- Missing citation: 26
- Missing citation: 27
- Missing citation: 28
- Missing citation: 29
- Missing citation: 30
- Missing citation: 31
- Missing citation: 32
- Missing citation: 33
- Missing citation: 34
- Missing citation: 35
- Missing citation: 36
- Missing citation: 37
- Missing citation: 38
- Missing citation: 39
- Missing citation: 40
- Missing citation: 41
- Missing citation: 42
- Missing citation: 43
- Missing citation: 44
- Missing citation: 45
- Missing citation: 46
- Missing citation: 47
- Missing citation: 48
- Missing citation: 49
- Missing citation: 50
- Missing citation: 51
- Missing citation: 52
- Missing citation: 53
- Missing citation: 54

- Missing citation: 55
- Missing citation: 62
- Missing citation: 63
- Missing citation: 64
- Missing citation: 65
- Missing citation: 66
- Missing citation: 67
- Missing citation: 68
- Missing citation: 69
- Missing citation: 70
- Missing citation: 71
- Missing citation: 72
- Missing citation: 73
- Missing citation: 74
- Missing citation: 75
- Missing citation: 76
- Missing citation: 77
- Missing citation: 78
- Missing citation: 79
- Missing citation: 80
- Missing citation: 81
- Missing citation: 82
- Missing citation: 83
- Missing citation: 84
- Missing citation: 85
- Missing citation: 86
- Missing citation: 87
- Missing citation: 88
- Missing citation: 89
- Missing citation: 90
- Missing citation: 91
- Missing citation: 92
- Missing citation: 93
- Missing citation: 94
- Missing citation: 95
- Missing citation: 96
- Missing citation: 97
- Missing citation: 98
- Missing citation: 99
- Missing citation: 100
- Missing citation: 108
- Missing citation: 109
- Missing citation: 110
- Missing citation: 111
- Missing citation: 112

- Missing citation: 113
- Missing citation: 114
- Missing citation: 115
- Missing citation: 116
- Missing citation: 117
- Missing citation: 104
- Missing citation: 105
- Missing citation: 106
- Missing citation: 118
- Missing citation: 119
- Missing citation: 120
- Missing citation: 121
- Missing citation: 122
- Missing citation: 123
- Missing citation: 124
- Missing citation: 125
- Missing citation: 126
- Missing citation: 127
- Missing citation: 128
- Missing citation: 129
- Missing citation: 130
- Missing citation: 131
- Missing citation: 132
- Missing citation: 133
- Missing citation: 134
- Missing citation: 135
- Missing citation: 136
- Missing citation: 137
- Missing citation: 138
- Missing citation: 139
- Missing citation: 140
- Missing citation: 141
- Missing citation: 142
- Missing citation: 143
- Missing citation: 144
- Missing citation: 145
- Missing citation: 146
- Missing citation: 147
- Missing citation: 148
- Missing citation: 149
- Missing citation: 150
- Missing citation: 151
- Missing citation: 152
- Uncited reference: 13
- Flagged reference: 3. Wa·sko, I.; Kozi·nska, A.; Kotlarska, E.; Baraniak, A. Clinically Relev

ant -Lactam Resistance Genes in Wastewater Treatment Plants. IJERPH 2022 , 19 , 13829. [ CrossRef ] [ PubMed ]
- Flagged reference: 7. Bush, K. Classication for -lactamases: Historical perspectives. Expert Rev. Anti-Infect. Ther. 2023 , 21 , 513 522. [ CrossRef ] [ PubMed ]

Suitability (4)
- Classified field: Environmental Microbiology / Antimicrobial Resistance Ecology
- Classified field: Water Science / Environmental Engineering and Wastewater Treatment
- Classified field: Public Health / One Health and Infectious Disease Epidemiology
- Classified field: Clinical Microbiology / Molecular Bacteriology of Beta-lactamases

Citation flags:
{"missing":["16","17","21","22","23","24","25","26","27","28","29","30","31","32","33","34","35","36","37","38","39","40","41","42","43","44","45","46","47","48","49","50","51","52","53","54","55","62","63","64","65","66","67","68","69","70","71","72","73","74","75","76","77","78","79","80","81","82","83","84","85","86","87","88","89","90","91","92","93","94","95","96","97","98","99","100","108","109","110","111","112","113","114","115","116","117","104","105","106","118","119","120","121","122","123","124","125","126","127","128","129","130","131","132","133","134","135","136","137","138","139","140","141","142","143","144","145","146","147","148","149","150","151","152"],"uncited":["13"],"duplicates":[],"status":"done","mode":"numeric-bracket","note":null,"reference_style":"numeric","reference_count":13,"list_truncated":true,"list_note":"The parsed reference list ends at 13, but the body cites up to [152]: the tail of the list was not parsed (merged entries or an unrecognised list format), so the 124 \"missing\" numbers above are unconfirmed, not proven absent.","crosscheck":null,"ambiguous":[]}

Fixes:
[must-fix] Re-parse full 1-471 reference list and re-run numeric-bracket check; do not report 16-152 as missing until tail is parsed
[must-fix] Split merged verification blocks into individual refs and re-verify; manually resolve flagged 3,7 and ambiguous 1,2,4,5,6,8,9,10,11,12
[must-fix] Remove false uncited 13; correct uncited list to empty pending re-parse
[must-fix] Temper WWTP reservoir/amplification and irrigation/drinking-water transmission claims to DNA-detection limits or add HGT, selection/co-selection, and persistence/expression/HGT framing; explicitly address viability vs extracellular DNA
[must-fix] Update or justify outdated citations 60 (2001) and 148 (2013) and 5 (2016) where used for current epidemiology/treatment efficacy; cite current systematic reviews for CTX-M/SHV/NDM/OXA
[nice-to-have] Drop irrelevant strengthening items Burnet 1959 and O’Donald 1962 and replace with AMR-relevant co-selection/mobilome literature; retain Gillings 2013 and Slizovskiy 2020 mobilome items
[nice-to-have] Clarify interdisciplinary scope as Environmental Microbiology/AMR Ecology + Water Engineering + One Health and verify Scopus indexing; follow Conclusions call to expand understudied regions and combine molecular with phenotypic persistence/expression/HGT work

MARVA does not guarantee acceptance at any journal.